NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

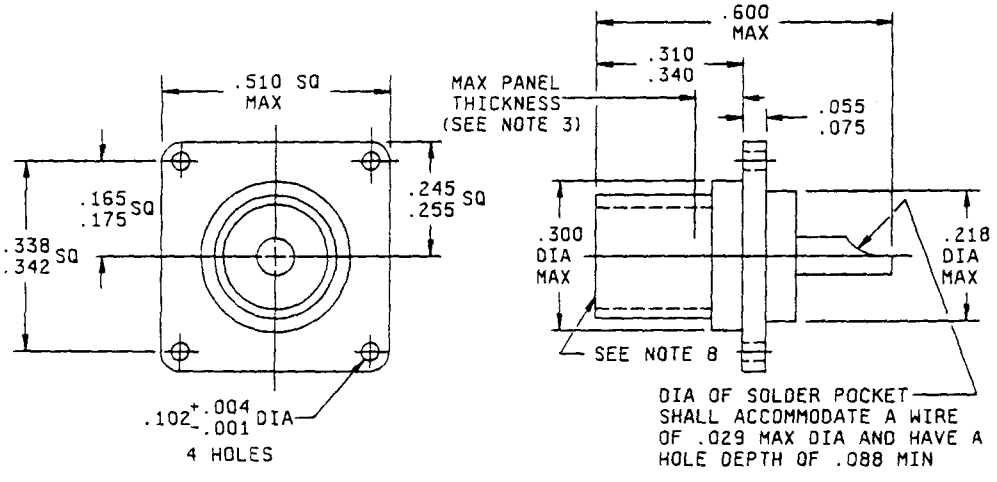
MIL-PRF-39012/60B
25 January 1995
SUPERSEDING
MIL-C-39012/60A
1 September 1984

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, RECEPTACLES, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SERIES SMA (UNCABLED, SOCKET CONTACT, FLANGE MOUNTED, CLASS 2)

This specification is approved for use by all Departments And Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-39012

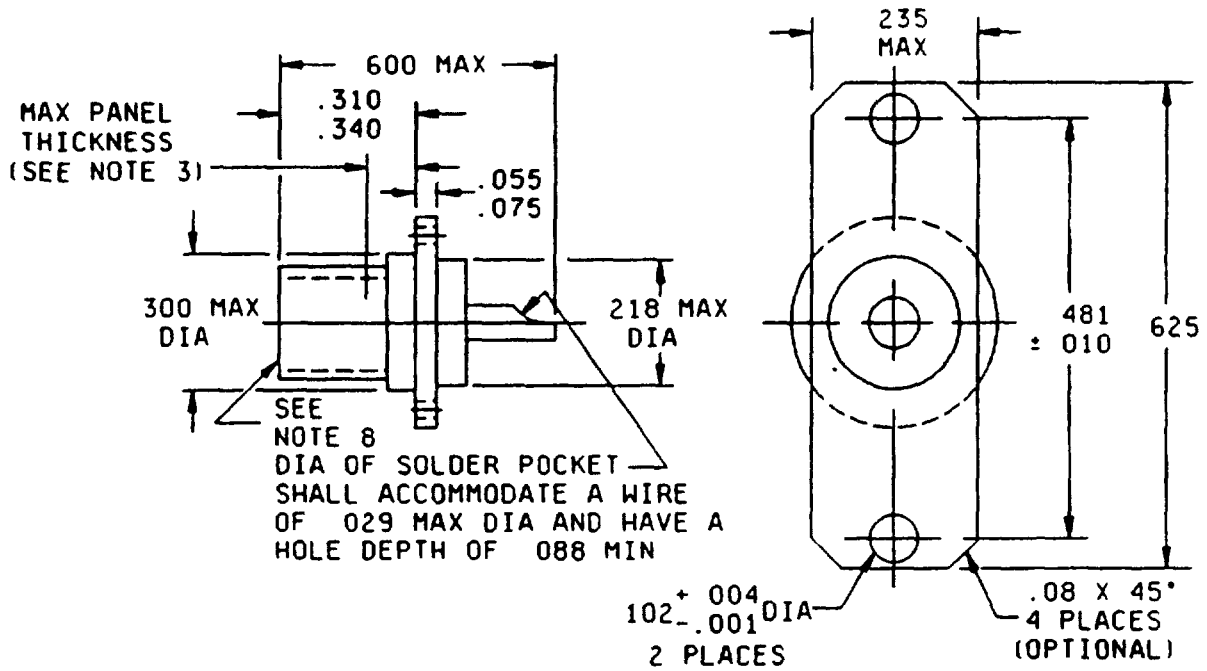


4-HOLE CONFIGURATION (M39012/60-3001 AND M39012/60-4001)

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.02	.08	2.0	.235	5.97	.340	8.64
.004	0.10	.088	2.24	.245	6.22	.342	8.69
.010	0.25	.102	2.59	.255	6.48	.481	12.22
.029	0.74	.165	4.19	.300	7.62	.510	19.25
.055	1.40	.175	4.45	.310	7.78	.600	15.24
.075	1.91	.218	5.54	.338	8.49	.625	15.88

FIGURE 1. General configuration.

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2-HOLE CONFIGURATION (M39012/60-3002 AND M39012/60-4002)

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Receptacle recommended for component part use and for use on panels with a .090 inch (2.29 mm) maximum thickness
4. All undimensioned pictorial configurations are for reference purposes only
5. There shall be a solid barrier in the socket between the pin entry and the solder pocket to prevent solder wicking
6. Unless otherwise specified, tolerance is ± .005 inch (0.13 mm) on three-place decimals, ± .010 inch (0.25 mm) on two-place decimals, and ±2° on angles
7. Solder pockets shall be gold plated in accordance with MIL-G-45204, type II, class I
8. Series SMA socket contact interface shall be in accordance with MIL-STD-348

FIGURE 1. General configuration - Continued

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ENGINEERING DATA

Nominal impedance 50 ohms

Frequency range Not rated

Voltage rating 335 volts rms maximum at sea level 85 volts rms maximum at 70 000 feet

Temperature rating 65°C to +165°C

REQUIREMENTS

Dimensions and configuration See figure 1 and MIL-STD-348

Force to engage and disengage

Longitudinal force Not applicable

Torque 2 inch-pounds maximum

Coupling proof torque Not applicable

Inspection conditions For each test of threaded coupling connector where the test is performed on mated pairs the pairs shall be torqued to 7 to 10 inch-pounds

Mating characteristics See MIL-STD 348 for dimensions

Contacts with spring members

Center contact (female)

Oversize test pin 0375 + 0001/- 0000 inch

Test pin finish 16 microinches

Insertion depth 030/ 045 inch

Number of insertions 3

Insertion force test

Steel test pin diameter 0370 + 0001/- 0000 inch

Insertion depth 050/ 075 inch

Test pin finish 16 microinches

Insertion force 2 pounds maximum

Withdrawal force test

Steel test pin diameter 0355 + 0000/- 0001 inch

Insertion depth 050/ 075 inch

Test pin finish 16 microinches

Withdrawal force 1 ounce minimum

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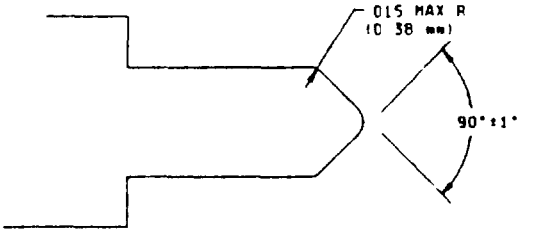


FIGURE 2 Test pin data

Hermetic seal Not applicable

Leakage (pressurized connectors) Not applicable

Insulation resistance Method 3003 MIL-STD-1344 5 000 megohms minimum

Center contact retention

Axial force 6 0 pounds minimum

Radial torque 4 inch-ounces minimum

Solderability 1/ Method 208 of MIL-STD-202

Resistance to test probe damage Not applicable

Corrosion (salt spray) Method 1001 test condition B MIL STD-1344

Voltage standing wave ratio (VSWR) Not applicable

Connector durability Method 2016 of MIL-STD-1344 500 cycles minimum at 12 cycles per minute maximum
The connector shall meet mating characteristics and force to engage and disengage requirements

Contact resistance (in milliohms maximum)

	Initial	After environment
Center contact	3 0	4 0
Outer contact	2 0	Not applicable

1/ For quality conformance inspection the test shall be performed in group B following the insulation resistance test

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Dielectric withstanding voltage at sea level Method 3001 of MIL-STD 1344 1 000 volts rms

Vibration high frequency Method 2005 test condition IV MIL-STD-1344

Shock Method 2004 test condition G MIL-STD-1344

Thermal shock Method 1003 test condition A MIL STD-1344 except test high temperature shall be +200°C

Humidity Method 1002 type II of MIL-STD-1344 No measurements at high humidity Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity

Corona level

  Voltage 250 volts minimum

  Altitude 70 000 feet

RF high potential withstanding voltage

  Voltage and frequency 670 volts rms at 5 MHz

  Leakage current Not applicable

Cable retention force Not applicable

Coupling mechanism retention force Not applicable

RF leakage Not applicable

RF insertion loss Not applicable

Group qualification See table I

TABLE I Group qualification and retention testing

Group	PINs 1/ 2/
I	M39012/60-3001 M39012/60-3002 M39012/61-3001 M39012/61-3002
II	M39012/60-4001 M39012/60-4002 M39012/61-4001 M39012/61-4002

1/ Submission and qualification of any connector in a group qualifies any and all connectors of the entire group

2/ For qualification retention data may be supplied on any part in a group to retain qualification for that entire group

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Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians
Army - CR
Navy - EC
Air Force - 85

Review activities
Army - AM AT ME MI
Navy - AS MC OS SH
Air Force - 19 99

Preparing activity
DLA - ES

(Project 5935-3911-01)

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